

Derivation of the Bilateral Teleoperation Algorithm

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Abstract

This formulation goes over the fundamental dynamical equations governing the multi degrees of freedom (DOF) series robotic manipulator and further specializes on the realization of several control algorithms such as bilateral control and/or embracing motion control. The derivation will be made on the basis of rigid body dynamics first on the joint space level and then on the task space level.

1. Joint Space Dynamics

The following derivation will assume the generalized coordinates of motion, preserving the flexibility of application for both translational and rotational joints of a robotic system. In that sense, the fundamental joint variable q will be used interchangeably and will refer to the rotation angle θ for a revolute joint and the joint displacement d for a prismatic joint. The same assumption will also be reflected on the first (i.e. $\dot{q}$ for $\dot{\theta}$ or $\dot{d}$) and second (i.e. $\ddot{q}$ for $\ddot{\theta}$ or $\ddot{d}$) derivatives of the generalized coordinate variable. Similarly, the input variable τ will refer to the force or the torque applied on a joint depending on whether that joint is revolute or prismatic. Finally, the terms mass (for prismatic) and inertia (for revolute) will be used interchangeably depending on the context.

The dynamics of a single-(DOF) mechanical system can be represented by the following mathematical model.

$$a(q)\ddot{q} + b(q, \dot{q}) + g(q) + \tau_{ext} = \tau_{in} \quad (1)$$

Here, the term $a_{min} \leq a(q) \leq a_{max}$ represents the bounded inertia of the system, $b_{min} \leq b(q, \dot{q}) \leq b_{max}$ stands for the coriolis, centripetal and friction forces with proper upper and lower bounds while the term $g_{min} \leq g(q) \leq g_{max}$ represent the bounded gravitational forces in the system. The term

$\tau_{e_{min}} \leq \tau_{ext} \leq \tau_{e_{max}}$ stands for the bounded external forces acting on the system. The expression given in (1) can be recast in the following form:

$$a(q)\ddot{q} = \tau_{in} - b(q, \dot{q}) - g(q) - \tau_{ext} \quad (2)$$

In practical applications, the term $a(q)$ can be characterized with a nominal value and a small variation around this nominal value as follows:

$$a(q) = a_n + \delta a \quad (3)$$

where a_n and δa respectively stand for the nominal inertia (or mass) of the corresponding link and the small variation around this nominal value. Now, combining this info with the expression given in (2), one can reformulate the dynamics of the system as follows:

$$a_n\ddot{q} + \delta a\ddot{q} = \tau_{in} - b(q, \dot{q}) - g(q) - \tau_{ext} \quad (4)$$

$$\ddot{q} = \frac{1}{a_n} \{ \tau_{in} - b(q, \dot{q}) - g(q) - \delta a\ddot{q} - \tau_{ext} \} \quad (5)$$

The right hand side of the structure shown in (5) can be divided into two parts: the input (i.e. the command) torque and the disturbances, which yields to the following:

$$\ddot{q} = \frac{1}{a_n} \{ \tau_{in} - \tau_{dis} \} \quad (6)$$

$$\tau_{dis} = b(q, \dot{q}) + g(q) + \delta a\ddot{q} + \tau_{ext} \quad (7)$$

Referring to the contents of (6) and (7), several control strategies can be employed:

- For a fully reconfigurable advanced motion control system the disturbances in (7) can be estimated using a Disturbance Observer which brings the system to an "acceleration controllable" form.
- For systems in which the external forces are either precisely known or equal to zero (such as industrial control systems), a proper system identification to estimate $b(q, \dot{q})$ and $\delta a\ddot{q}$ can be made and an inverse dynamics control structure can be employed.
- For some reconfigurable robotic systems, internally embedded gravity compensation algorithms and joint force/torque measurement or estimation interfaces may exist.

The system in the **Cobot Maker Space** can basically be categorized under the last option. The system accepts input torque commands (τ_{in}), and can apply dynamical gravity compensation (i.e. $g(q) \approx 0$). Furthermore, the torque content due to the variation of inertia ($\delta a \ddot{q}$) and the coriolis, centripetal and friction terms ($b(q, \dot{q})$) can be compensated using a Disturbance Observer (DOB). Therefore, for this system, the equation given in (5)

$$\ddot{q} = \frac{1}{a_n} \left\{ \tau_{in} - \cancel{b(q, \dot{q})} \overset{\approx 0}{\rightarrow} - \cancel{g(q)} \overset{\approx 0}{\rightarrow} - \cancel{\delta a \ddot{q}} \overset{\approx 0}{\rightarrow} - \tau_{ext} \right\} \quad (8)$$

$$\ddot{q} = \frac{1}{a_n} \{ \tau_{in} - \tau_{ext} \} \quad (9)$$

So, for the system under consideration, the main goal in the control design is to acquire and provide the desired accelerations in the real-time control action. The following section makes this formulation for the bilateral teleoperation algorithm.

1.1. Disturbance Observer

For the dynamic system given in (7), removing the non modeled (i.e. disturbance) torque (τ_{dis}) is of crucial importance. As this disturbance term contains all undesired torques, in case that it is removed from the system, one can come up with a double integrator structure. In that structure it will be possible to give the acceleration reference of any motion to have it done. Such a system is called an acceleration controlled plant and the control algorithm is usually referred as the acceleration control algorithm. In order to estimate and reject the disturbance acting to the system, a disturbance observer (DOB) can be realized. One way to interpret the disturbance torque is to observe the difference between what is given to the system and what the system outputs. In other words, the difference between the input torque to the joint (i.e. τ_{in}) and the joint's response torque (i.e. $a_n \ddot{q}$) simply represents the disturbance torque appearing in the system. However, direct subtraction of the response torque from that of the input torque would yield to a signal that does not have enough phase shift to be used in a feedback loop. In order to provide a small phase shift for the feedback (i.e. lag control structure) and potentially attenuate the noise in the system, one can use a low-pass filter. Switching to frequency domain, this could be achieved as:

$$T_{dis}(s) = T_{in}(s) - a_n s^2 Q(s) \quad (10)$$

$$\hat{T}_{dis}(s) = \{T_{in}(s) - a_n s^2 Q(s)\} \left(\frac{g}{s+g} \right) \quad (11)$$

Now, without loss of generality, one can slightly shuffle the expression given in (11) to include feedback from the velocity measurement (or estimation) as follows:

$$\hat{T}_{dis}(s) = T_{in}(s) \left(\frac{g}{s+g} \right) - a_n s Q(s) \left(\frac{sg}{s+g} \right) \quad (12)$$

$$\hat{T}_{dis}(s) = T_{in}(s) \left(\frac{g}{s+g} \right) - a_n \Omega(s) \left(\frac{sg}{s+g} \right) \quad (13)$$

$$\hat{T}_{dis}(s) = T_{in}(s) \left(\frac{g}{s+g} \right) - a_n g \Omega(s) \left(1 - \frac{g}{s+g} \right) \quad (14)$$

$$\hat{T}_{dis}(s) = \{T_{in}(s) + a_n g \Omega(s)\} \left(\frac{g}{s+g} \right) - a_n g \Omega(s) \quad (15)$$

Here, selection of the observer cut-off frequency " g " is of crucial importance. Theoretically, this parameter should be selected as high as possible. However, due to the limits imposed by the discrete real-time sampling of the system (rooted back to the Shannon's sampling theorem), this selection is limited. Empirical values for a system running at 1 KHz loop frequency is usually in the range $500 < g < 2500$.

2. Interaction Force Model

To design the force-control loop of a bilateral controller, it is necessary to characterize the physical properties of the environment that the master or slave device interacts with. Various studies have proposed mathematical models for representing the interaction force. Among these, the most widely adopted approach is the standard contact-impedance model, expressed as:

$$f_e = \delta m_e (\ddot{x} - \ddot{x}_e) + b_e (\dot{x} - \dot{x}_e) + k_e (x - x_e) \quad (16)$$

where, the terms δm_e , b_e and k_e respectively stand for the deflected environment mass, damping and the stiffness of the environment. In (16), variables x and x_e (for $x \geq x_e$) represent the position response of the corresponding system and the environment contact position in the same cartesian reference frame. In most of the practical scenarios, the environment location in that

reference frame can be assumed to have a static structure which results in $\ddot{x} = 0$ and $\dot{x} = 0$. This assumption simplifies the expression in (16) as follows:

$$f_e = \delta m_e \ddot{x} + b_e \dot{x} + k_e(x - x_e) \quad (17)$$

In this model, the interaction force arises from three primary components: inertial forces ($\delta m_e \ddot{x}$) which result from accelerating the mass being moved; viscous forces ($b_e \dot{x}$) which stem from motion through a resistive medium; and elastic or spring forces ($k_e(x - x_e)$), which are produced when the environment is displaced from its equilibrium position. Examining these components more closely reveals the typical relationship among most physical systems as $\|k_e\| \gg \|b_e\| \gg \|\delta m_e\|$. This implies that the stiffness term dominates the overall interaction force. Put differently, this approximation allows one to neglect transient dynamics and instead emphasize the steady-state interaction force during force control. Consequently, many studies in robotic force-control applications simplify the environment model to the following spring dominated form:

$$f_e = k_e(x - x_e) \quad (18)$$

Furthermore, under the assumption of a static environment, the derivatives of the interaction force can be written as follows:

$$\dot{f}_e = k_e \dot{x} \quad (19)$$

$$\ddot{f}_e = k_e \ddot{x} \quad (20)$$

Without loss of generality, one can also write similar equations for a system consisted of revolute joint(s) as follows:

$$\tau_e = k_e(q - q_e) \quad (21)$$

$$\dot{\tau}_e = k_e \dot{q} \quad (22)$$

$$\ddot{\tau}_e = k_e \ddot{q} \quad (23)$$

3. Derivation of the Teleoperation Controller

The goal of the controller design process is to come up with the necessary (i.e. desired) accelerations for the master and slave systems that would enforce the dynamics such that the position and force tracking errors move to zero. In the following subsections, the acceleration relationships for the position and force control channels in the joint space domain will be obtained.

The last subsection is reserved for the common solution to obtain independent desired accelerations for the master and the slave systems. Since the goals of a bilateral control system is to synchronize the positions and the interaction forces, one can define the following tracking errors:

$$e_q = q_m - q_s \quad (24)$$

$$e_\tau = \tau_m + \tau_s \quad (25)$$

where the terms q_m , q_s , τ_m and τ_s represent the master position response, slave position response, master torque response and slave torque response all in the configuration (i.e. joint) space, respectively.

3.1. Position Control Loop

The enforcement of the position tracking error from any initial value to zero can be made using the following error dynamics:

$$\ddot{e}_q + C_v \dot{e}_q + C_q e_q = 0 \quad (26)$$

Provided that the control coefficients are strictly positive (i.e. $C_v > 0$ and $C_q > 0$), expression given in (26) results in a convergent solution which forces $e_q \xrightarrow[t \rightarrow \infty]{} 0$. Besides the stable convergence condition, referring to the second order ODE dynamics, selection of $C_v^2 - 4C_q \geq 0$ provides a transient without any oscillations. As long as this condition is satisfied, the control designer has to pick the highest possible positive values for C_v and C_q for the fastest convergence of error to zero.

Now, by making proper substitution from (24) to (26), one can obtain the following expressions:

$$(\ddot{q}_m - \ddot{q}_s) + C_v (\dot{q}_m - \dot{q}_s) + C_q (q_m - q_s) = 0 \quad (27)$$

$$\ddot{q}_m - \ddot{q}_s = -C_v (\dot{q}_m - \dot{q}_s) - C_q (q_m - q_s) \quad (28)$$

3.2. Force Control Loop

A similar approach to that of the position control channel can be implemented for the force control error dynamics as follows:

$$\ddot{e}_\tau + C_y \dot{e}_\tau + C_f e_\tau = 0 \quad (29)$$

Assuming the control coefficients are strictly positive (i.e., $C_y > 0$ and $C_f > 0$), the expression in (29) yields a convergent response that drives $e_\tau \xrightarrow[t \rightarrow \infty]{} 0$.

0. Similar to the position loop, in addition to ensuring stable convergence, selecting $C_y^2 - 4C_f \geq 0$ based on the second-order ODE dynamics guarantees a transient behavior without oscillations.

Substituting (25) to (29), one can obtain the following dynamical equation:

$$(\ddot{\tau}_m + \ddot{\tau}_s) + C_y (\dot{\tau}_m + \dot{\tau}_s) + C_f (\tau_m + \tau_s) = 0 \quad (30)$$

Making use of (22) and (23), one can rearrange (30) to obtain either one of the following two equations:

$$k_m \ddot{q}_m + k_s \ddot{q}_s = -C_y (\dot{\tau}_m + \dot{\tau}_s) - C_f (\tau_m + \tau_s) \quad (31)$$

$$k_m \ddot{q}_m + k_s \ddot{q}_s = -C_y (k_m \dot{q}_m + k_s \dot{q}_s) - C_f (\tau_m + \tau_s) \quad (32)$$

Among those expressions, the first one (i.e. (31)) contains the derivative of the torques and the second one (Eq. (32)) makes use of the spring model and velocity feedback. Depending on the availability of the proper signals (such as the derivative of the force), the designer can pick anyone of these two equations. For the Panda robots in the [Cobot Maker Space](#), selection of equation (32) is a better option since force signal from the sensors might contain noisy information preventing differentiation with an acceptable signal-to-noise ratio (SNR).

3.3. Combined Teleoperation Controller

Equations (28) and (32) contain both of the master and slave desired accelerations simultaneously satisfying error convergence for both of the position and force channels. The remaining task for this subsection is to combine these two equation sets and decouple the master and the slave desired accelerations. For this purpose, one can modify these two equations and combine them in the following matrix equality:

$$\begin{bmatrix} 1 & -1 \\ k_m & k_s \end{bmatrix} \begin{bmatrix} \ddot{q}_m \\ \ddot{q}_s \end{bmatrix} = \begin{bmatrix} -C_q & C_q & -C_v & C_v & 0 & 0 \\ 0 & 0 & -C_y k_m & -C_y k_s & -C_f & -C_f \end{bmatrix} \begin{bmatrix} q_m \\ q_s \\ \dot{q}_m \\ \dot{q}_s \\ \tau_m \\ \tau_s \end{bmatrix} \quad (33)$$

Since both of the master and slave system stiffness values are positive, the matrix in front of the accelerations is always invertible. Using this information, one can now write down the desired accelerations for the master and

the slave systems as follows:

$$\begin{bmatrix} \ddot{q}_m^{des} \\ \ddot{q}_s^{des} \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ k_m & k_s \end{bmatrix}^{-1} \begin{bmatrix} -C_q & C_q & -C_v & C_v & 0 & 0 \\ 0 & 0 & -C_y k_m & -C_y k_s & -C_f & -C_f \end{bmatrix} \begin{bmatrix} q_m \\ q_s \\ \dot{q}_m \\ \dot{q}_s \\ \tau_m \\ \tau_s \end{bmatrix} \quad (34)$$

Carrying out the necessary algebra, one can write down the desired accelerations as follows:

$$\ddot{q}_m^{des} = \frac{1}{k_m + k_s} \{ -C_q k_s q_m + C_q k_s q_s - (C_y k_m + C_v k_s) \dot{q}_m + (C_v k_s - C_y k_s) \dot{q}_s - C_f \tau_m - C_f \tau_s \} \quad (35)$$

$$\ddot{q}_s^{des} = \frac{1}{k_m + k_s} \{ C_q k_m q_m - C_q k_m q_s + (C_v k_m - C_y k_m) \dot{q}_m - (C_v k_m + C_y k_s) \dot{q}_s - C_f \tau_m - C_f \tau_s \} \quad (36)$$

In (35) and (36), the master and the slave side environment stiffness values (i.e. k_m and k_s) act as controller design parameters. In other words, the real values of those parameters could easily be modified by properly adjusting the remaining controller coefficients which would yield the same controller dynamics. So, without loss of generality, one can assume them to be equal to a constant value as:

$$k_m = k_s = K \quad (37)$$

Substituting (37) back to (35) and (36) results in:

$$\ddot{q}_m^{des} = -\frac{C_q}{2}(q_m - q_s) - \frac{C_v}{2}(\dot{q}_m - \dot{q}_s) - \frac{C_y}{2}(\dot{q}_m + \dot{q}_s) - \frac{C_f}{2K}(\tau_m + \tau_s) \quad (38)$$

$$\ddot{q}_s^{des} = -\frac{C_q}{2}(q_s - q_m) - \frac{C_v}{2}(\dot{q}_s - \dot{q}_m) - \frac{C_y}{2}(\dot{q}_m + \dot{q}_s) - \frac{C_f}{2K}(\tau_m + \tau_s) \quad (39)$$

4. Harmonic Gearbox Dynamics

The harmonic gearbox between the motor and the link side provides certain compliance to the motion of the system. However, this compliance negatively affects the force to be perceived on the motor side of the corresponding joint. The Franka Panda robot measures the external force via torque sensors at the link side. The mapping of those forces to the motor side, which would otherwise be done by a simple scaling, can only be made if an approximate model (i.e. a transfer function) for the harmonic gearbox could be obtained.

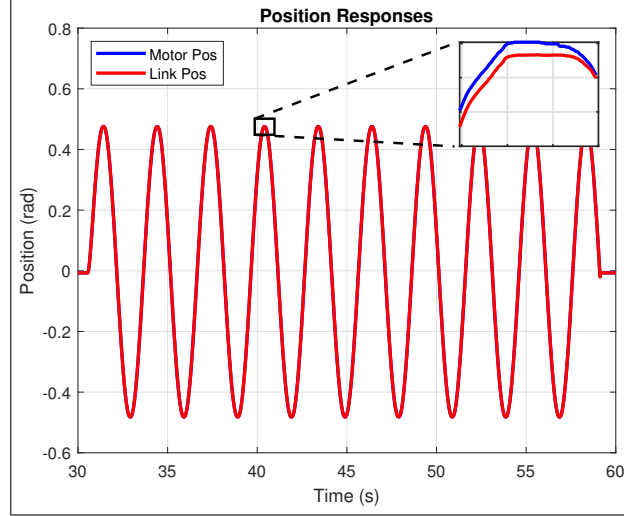


Figure 1: Motor and Link Position Responses

For this purpose, we conducted a series of experiments and recorded the respective motor side position $q_m(t)$ and the link side position $q_l(t)$ as shown below in Figure. 1. Assuming a standard impedance model for the harmonic drive which is consisted of a mass-spring-damper structure. The following equations show a step-by-step derivation of this transfer function.

The torque transferred through the flexible flexspline is expended in accelerating the load inertia and counteracting viscous damping.

$$\tau_f(t) = J_l \ddot{q}_l(t) \quad (40)$$

where, $\tau_f(t)$ and J_l stand for the torques applied by the flexible structure of the harmonic gearbox and the inertia of the link, respectively. The same transmitted gearbox torque ($\tau_f(t)$) is determined by the elastic and dissipative characteristics of the transmission, represented by the stiffness K and damping D , respectively. These quantities act on the twist angle, which is defined as the relative angular displacement between the input position reflected to the output side via the reduction ratio N and the measured output position.

$$\tau_f(t) = K \left(\frac{q_m(t)}{N} - q_l(t) \right) + D \left(\frac{\dot{q}_m(t)}{N} - \dot{q}_l(t) \right) \quad (41)$$

Combining the identities in (40) and (41) results in the following expression:

$$J_l \ddot{q}_l(t) = K \left(\frac{q_m(t)}{N} - q_l(t) \right) + D \left(\frac{\dot{q}_m(t)}{N} - \dot{q}_l(t) \right) \quad (42)$$

Taking the Laplace transform of (42) yield to:

$$J_l s^2 Q_l(s) = K \left(\frac{Q_m(s)}{N} - Q_l(s) \right) + Ds \left(\frac{Q_m(s)}{N} - Q_l(s) \right) \quad (43)$$

Grouping $Q_l(s)$ and $Q_m(s)$, one can obtain:

$$(J_l s^2 + Ds + K) Q_l(s) = \left(\frac{Ds + K}{N} \right) Q_m(s) \quad (44)$$

With a final touch, this last expression can be rearranged to yield the motor to link transfer function as follows:

$$\frac{Q_l(s)}{Q_m(s)} = \frac{Ds + K}{N (J_l s^2 + Ds + K)} \quad (45)$$

$$\frac{Q_l(s)}{Q_m(s)} = \frac{Ds + K}{(N J_l s^2 + N Ds + N K)} \quad (46)$$

This expression could be normalized to result in a monic polynomial at the denominator:

$$\frac{Q_l(s)}{Q_m(s)} = \frac{\alpha s + \beta}{(s^2 + \gamma s + \delta)} \quad (47)$$

where, $\alpha = \frac{D}{N J_l}$, $\beta = \frac{K}{N J_l}$, $\gamma = \frac{D}{J_l}$ and $\delta = \frac{K}{J_l}$.

4.1. Estimation of Coefficients

The transfer function given in (47) represents a continuous-time approximation of the harmonic gearbox dynamics. Although this expression can be converted to the discrete-time domain and rearranged into a linear regression form, such an equation-error based least-squares formulation may lead to biased parameter estimates when the measured output signal contains noise. This issue stems from the fact that the link-side position measurement appears both in the regression matrix and in the output vector of the least-squares problem. Furthermore, the finite-difference operations required by such a formulation amplify the measurement noise and may result in physically inconsistent parameter estimates. For this reason, instead of solving

a direct pseudo-inverse based least-squares problem, an output-error based nonlinear least-squares formulation is adopted. Let the experimentally measured motor-side and link-side position signals be denoted by $q_m[k]$ and $q_l[k]$, respectively. For a given parameter vector

$$\boldsymbol{\theta} = [\alpha \quad \beta \quad \gamma \quad \delta]^T \quad (48)$$

the continuous-time harmonic gearbox model can be written as

$$G(s, \boldsymbol{\theta}) = \frac{Q_l(s)}{Q_m(s)} = \frac{\alpha s + \beta}{s^2 + \gamma s + \delta} \quad (49)$$

The model given above is first converted to its discrete-time equivalent using the sampling period of the real-time control system. Subsequently, the measured motor-side position signal is applied to the discrete-time model in order to generate the simulated link-side position response

$$\hat{q}_l[k; \boldsymbol{\theta}] = G_d(z, \boldsymbol{\theta}) q_m[k] \quad (50)$$

where, $G_d(z, \boldsymbol{\theta})$ denotes the discrete-time equivalent of the transfer function given in (49). The identification problem is then formulated as the minimization of the difference between the measured and simulated link-side position signals. Accordingly, the optimal parameter vector is obtained as:

$$\boldsymbol{\theta}^* = \arg \min_{\boldsymbol{\theta}} \sum_{k=0}^n (q_l[k] - \hat{q}_l[k; \boldsymbol{\theta}])^2 \quad (51)$$

It should be noted that the coefficients in (49) are not estimated through a direct pseudo-inverse solution. Although such a solution is algebraically straightforward, it does not impose the physical constraints of the harmonic gearbox model and may yield negative damping- or stiffness-related coefficients. Therefore, a constrained nonlinear optimization approach is preferred in this study. In order to preserve the physical meaning of the identified gearbox model, the optimization is carried out under the positivity constraints as:

$$\alpha > 0, \quad \beta > 0, \quad \gamma > 0, \quad \delta > 0 \quad (52)$$

These constraints ensure a stable second-order denominator and a minimum-phase numerator.

5. Experimental Validation

The algorithm given above is tested over a set of experiments in **one of the joints** of the Franka Panda robot. In these experiments, the two versions of the joint position (i.e. θ and q with LibFranka notation) respectively standing for the motor side and the link side measurements are used. It should be noted that the position readouts for the motor side of the corresponding links are given with proper scaling of the gear ratio. Hence the difference between these two measurements are purely due to the dynamics enforced by the harmonic gearbox in between.

To estimate the dynamics of the harmonic gearbox, we had a total of 9 experiments. Among those experiments, the first 4 ones were of standard sinusoidal type with various amplitudes and frequencies, while the last 4 data consisted of pulse signals with varying amplitudes and frequencies. The 5th data, on the other hand, contains a sine-sweep between 0 and 10 Hz to observe the effects of intermediate frequencies. A concatenated image of the recorded data from these 9 experiments are shown below in Figure 2.

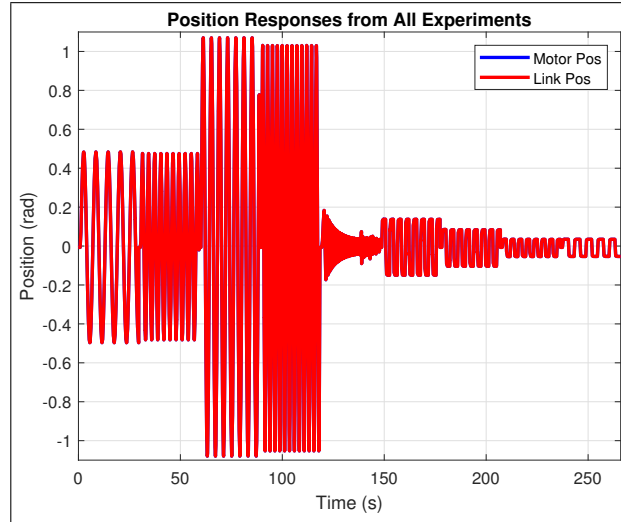


Figure 2: Concatenated Experiments

The experiment set shown in Figure 2 is used for the estimation of the coefficients of the transfer function (47). The non-linear estimation described in the previous section, specifically the constrained optimization shown in (51). The implementation was made using the entire data set collected from all

experiments. The convergence is achieved at 134 steps using the *lsqnonlin()* function of Matlab. The converged parameters have the following values:

$$\alpha = 9.5834e + 05 \quad (53)$$

$$\beta = 1.0936e + 07 \quad (54)$$

$$\gamma = 9.6366e + 05 \quad (55)$$

$$\delta = 1.0936e + 07 \quad (56)$$

The experimental identification results revealed that the estimated coefficients satisfy the relationships

$$\alpha \approx \gamma, \quad \beta \approx \delta \quad (57)$$

Referring to the definitions of the coefficients given below (47), these results indicate that the reduction ratio does not explicitly appear as an observable scaling factor in the recorded measurements as also expected from the graphical inspection of the recorded position data. In other words, the motor-side and link-side position signals provided by the robot are already represented in compatible coordinates. Consequently, the identified transfer function becomes consistent with the simplified structure

$$\frac{Q_l(s)}{Q_m(s)} \approx \frac{\alpha s + \beta}{s^2 + \alpha s + \beta} \quad (58)$$

Finally, the identified transfer function was validated experimentally by applying the measured motor-side position trajectory to the identified model and comparing the resulting simulated link-side position response against the actual measured link-side position. The obtained results demonstrated in Figure 3 show a significant reduction in the prediction error, confirming that the identified model captures the dominant dynamics of the harmonic gearbox with sufficient accuracy for control design purposes.

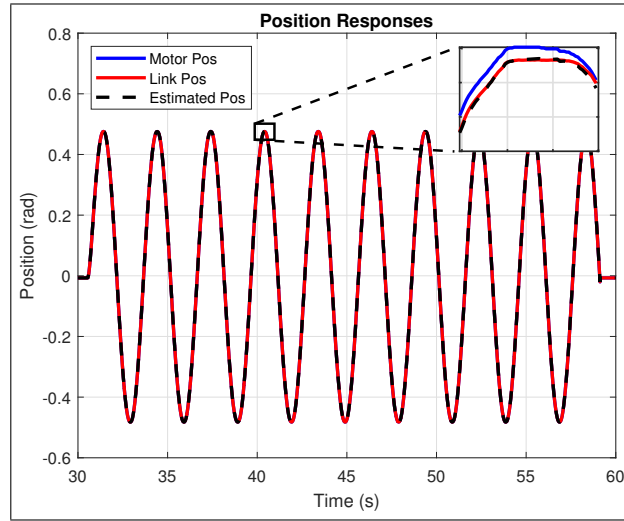


Figure 3: Experiments Reproduced With The Estimated Transfer Function

Further validation is obtained by examining the estimation errors. Figure 4 shows the errors between the recorded motor-side and link-side positions and the estimated and actual link positions.

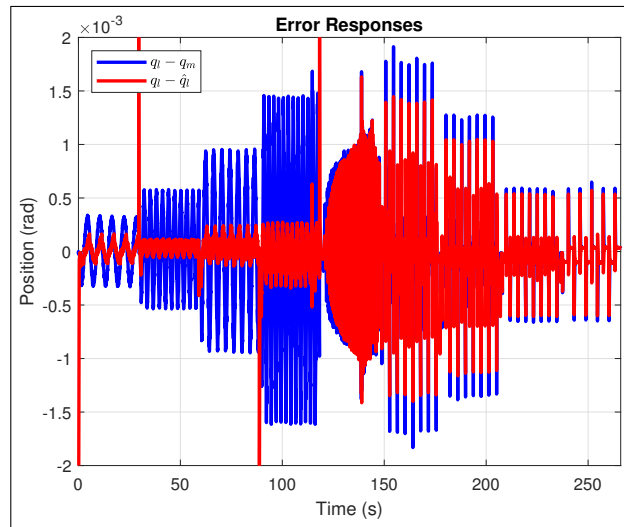


Figure 4: Comparison of Errors